

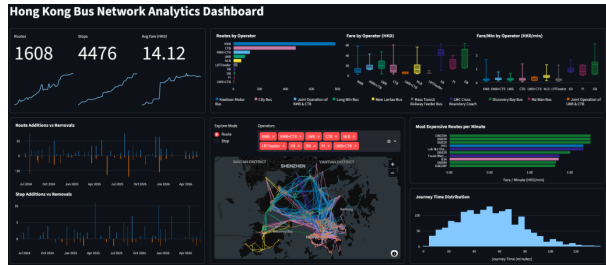
DATA VISUALIZATION PROJECT

Hong Kong Bus Network Analytics Dashboard

Course: Data Visualization

Class: 168068

Instructor: Dr. Tran Viet Trung



Project Members

Student ID	Full Name
20235474	Tran Nguyen My Anh
20235533	Nguyen Ngoc Minh
20235483	Tran Le Ha Dan
20235571	Kieu Son Tung

1. Visualization Problem and Design Goal

Problem

Raw transport records are relational and technical. They contain routes, stops, fares, companies, and time snapshots, but they do not immediately reveal network patterns.

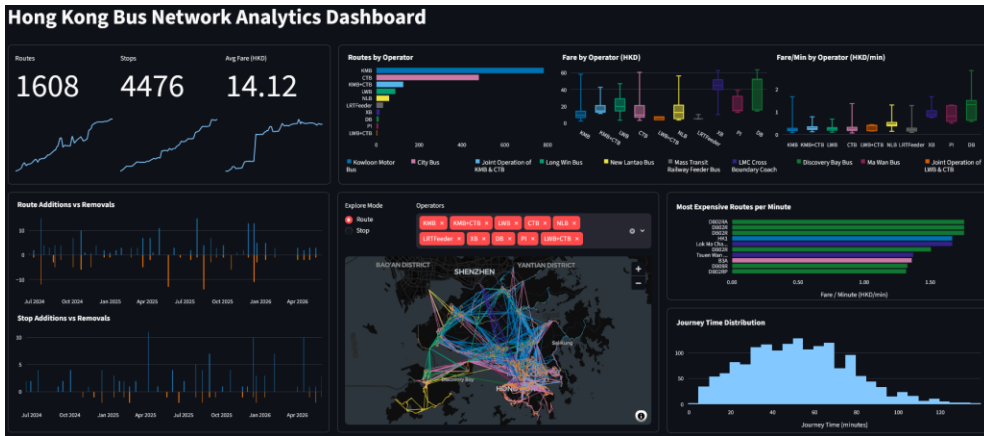
Target audience

Urban planners and transport policy analysts who need fast overview plus detailed exploration.

Data visualization goal

- ▶ Turn tables into marks, channels, and coordinated views.
- ▶ Support overview first, then drill-down.
- ▶ Make comparison possible without reading raw rows.
- ▶ Preserve visual integrity and accessibility.

2. Dashboard as a Visual Analytics Interface



Design type

Explanatory summary combined with exploratory interaction.

Reading path

KPI summary → operator comparison → temporal change → spatial network → outlier diagnostics.

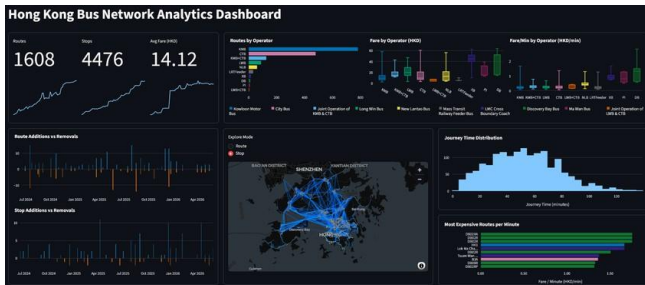
3. From Data Tables to Visual Variables

Dataset scope

Official DATA.GOV.HK "Routes and Fares of Public Transport" dataset; 55 snapshots from 1 June 2024 to 30 May 2026.

Attribute	Data type	Visual encoding in the dashboard
Company code	Nominal	Stable hue across charts and map.
Counts, fare, time	Quantitative	Position/length for comparison; distributions for spread.
Snapshot date	Temporal	X-position in sparklines and timelines.
Stop location	Spatial	Geographic map position.
Degree / density	Quantitative	Point size, line density, and tooltip detail.

4. Information Architecture and Storytelling



Beginning

KPI cards give immediate system status.

Middle

Operator, fare, and temporal panels explain variation across categories and time.

End

Map and anomaly panels support investigation of spatial clusters and unusual routes.

Design principle: overview-to-detail narrative, not isolated charts.

5. KPI Cards and Sparklines



Why cards?

Large numeric text is the clearest mark for exact system status: routes, stops, and average fare.

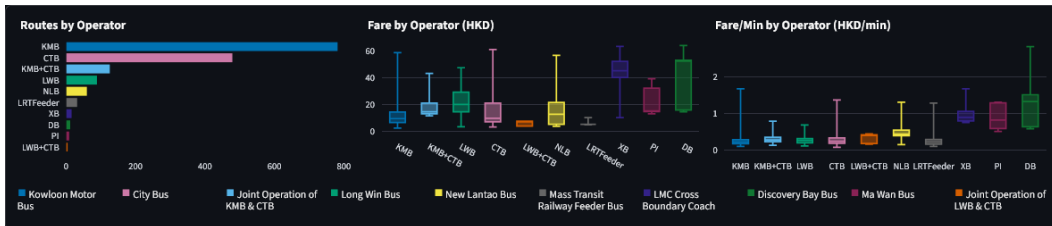
Why sparklines?

Small line charts add trend context without consuming a full chart area.

Visual principle

Proximity groups the KPIs as one summary; direct labels reduce lookup effort.

6. Operator Comparison: Bar Length and Box Plots



Horizontal bar chart

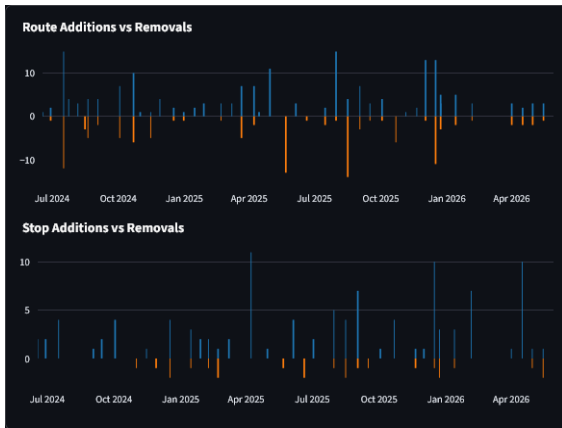
- ▶ Task: rank operators by route share.
- ▶ Reason: length is accurate for amount comparison.

Box plot

- ▶ Task: compare median, spread, and outliers.
- ▶ Reason: avoids hiding variation behind a mean.

Consistent operator colors support pre-attentive grouping across views.

7. Temporal Dynamics: Diverging Change Charts



Visual encoding

Additions are drawn above a zero baseline; removals are drawn below it.

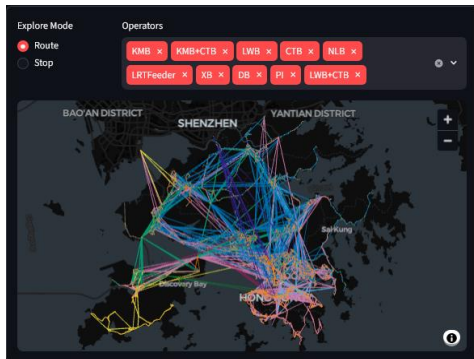
Why this chart?

A diverging bar chart makes growth and contraction comparable in one temporal frame.

Course principle

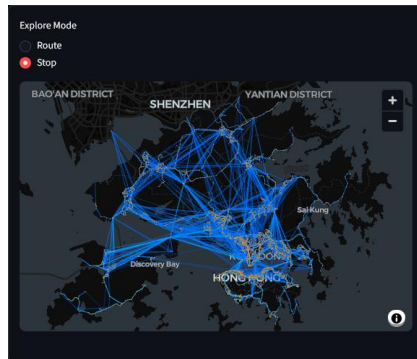
Position relative to a shared baseline makes sign and magnitude easier to judge than separate tables.

8. Spatial Network Map: Points, Lines, and Geography



Graph representation

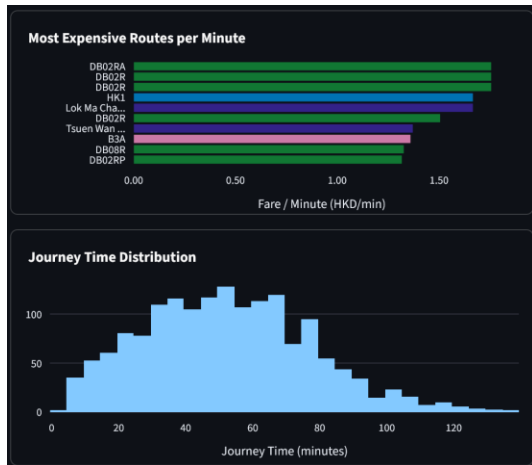
Stops are nodes; consecutive stop pairs are edges.



Channels

Geographic position encodes location; line color encodes operator; point size reflects stop centrality.

9. Distribution and Outlier Views



Histogram

Bins journey times to show the overall shape of service duration across routes.

Ranked bar chart

Sorts fare-per-minute values so expensive routes become visible quickly.

Visual role

These views move the dashboard from monitoring to diagnostic analysis.

10. Interaction Design

Filter

Operator selections update the network view and reduce visual density.

Mode switch

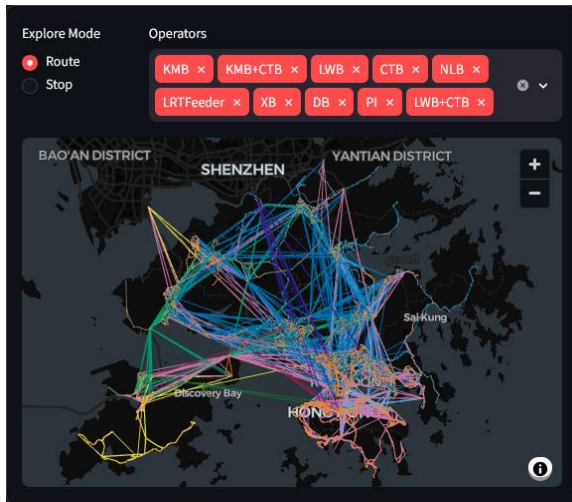
Route mode emphasizes flows; Stop mode emphasizes hubs and stop-level density.

Zoom and pan

Geographic interaction supports details-on-demand while preserving city-scale context.

Tooltips

Hover details provide exact values without cluttering charts with labels.



Design rationale: explanatory structure gives the user a path; interaction gives them control over local detail.

11. Color, Gestalt, and Visual Integrity

Color

- ▶ Okabe–Ito colorblind-safe palette.
- ▶ Operator colors remain stable across panels.
- ▶ Color is used for categories, not exact quantities.

Gestalt

Proximity groups related panels; similarity links repeated operator colors.

Okabe & Ito



Visual integrity

- ▶ No 3D effects.
- ▶ Direct labels reduce legend dependence.
- ▶ High data-ink ratio keeps the dashboard readable.

12. Design Trade-offs and Possible Improvements

Dense network map

Full route rendering can be visually crowded and computationally heavy.

Many categories

Ten operator colors approach the limit of reliable hue discrimination.

Widget limitation

Streamlit multiselect labels cannot show color swatches matching operator colors.

Hong Kong Bus Network Analytics Dashboard

Thanks for Listening

Questions and Live Dashboard Demonstration

Hong Kong Bus Network Analytics Dashboard

